

Name: _____

Date: _____

TRIGONOMETRY, (SOH CAH TOA FINDING SIDES)

LO: I can use trigonometric ratios to find unknown side lengths in right-angled triangles.

Finding sides with trigonometry

To find an **unknown side**, label the sides (O, A, H) relative to the given angle, then use the appropriate ratio: **SOH, CAH, or TOA**.

Rearrange: If the unknown is in the denominator, rearrange the formula or use the triangle method.

Quick examples

●	hyp = 10, angle = 30°: opp = $10 \sin 30^\circ = 5$	multiply hyp by sin
●	adj = 6, angle = 45°: opp = 6 $\tan 45^\circ = 6$	multiply adj by tan
●	opp = 8, angle = 30°: hyp = $8 \div \sin 30^\circ = 16$	divide by sin
●	opp = hyp $\div \sin \theta$	opp = hyp $\times \sin \theta$

Key idea

Label sides O, A, H. Choose the ratio involving the known side and the unknown side. Rearrange to find the unknown.

$$\text{opp} = \text{hyp} \times \sin \theta$$

Similarly: $\text{adj} = \text{hyp} \times \cos \theta$, $\text{opp} = \text{adj} \times \tan \theta$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding the opposite

Angle = 30°, hypotenuse = 12. Find the opposite.

Need opp, have hyp and angle: use sin

$$\sin 30^\circ = \text{opp}/12$$

$$\text{opp} = 12 \times \sin 30^\circ = 12 \times 0.5 = 6$$

$$\text{opp} = 6$$

Multiply the hypotenuse by sin of the angle.

Example 2 - finding the adjacent

Angle = 60°, hypotenuse = 10. Find the adjacent.

Need adj, have hyp: use cos

$$\cos 60^\circ = \text{adj}/10$$

$$\text{adj} = 10 \times \cos 60^\circ = 10 \times 0.5 = 5$$

$$\text{adj} = 5$$

Multiply hypotenuse by cos.

Example 3 - finding the hypotenuse

Angle = 45°, opposite = 7. Find the hypotenuse.

$$\sin 45^\circ = 7/\text{hyp}$$

$$\text{hyp} = 7 \div \sin 45^\circ$$

$$\text{hyp} = 7 \div \left(\frac{1}{\sqrt{2}}\right) = 7 \times \sqrt{2} = 9.9 \text{ (1 dp)}$$

$$\text{hyp} = 7 \times \sqrt{2} = 9.9$$

When the unknown is the denominator, divide.

Example 4 - using tan

Angle = 40°, adjacent = 8. Find the opposite (1 dp).

$$\tan 40^\circ = \text{opp}/8$$

$$\text{opp} = 8 \times \tan 40^\circ$$

$$= 8 \times 0.839 = 6.7$$

$$\text{opp} = 6.7$$

Use tan when you have opposite and adjacent.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding opposite (using sin)

Find the opposite side.

- a) hyp = 10, angle = 30°
- b) hyp = 8, angle = 45°
- c) hyp = 14, angle = 60°
- d) hyp = 20, angle = 30°

2. Finding adjacent (using cos)

Find the adjacent side.

- a) hyp = 12, angle = 60°
- b) hyp = 16, angle = 45°
- c) hyp = 6, angle = 30°
- d) hyp = 24, angle = 60°

3. Finding hypotenuse

Find the hypotenuse.

- a) opp = 4, angle = 30°
- b) adj = 5, angle = 60°
- c) opp = 6, angle = 45°
- d) adj = 9, angle = 30°

4. Using tan

Find the unknown side using tan.

- a) adj = 5, angle = 45°. Find opp.
- b) opp = 8, angle = 45°. Find adj.
- c) adj = 4, angle = 60°. Find opp.
- d) opp = 10, angle = 30°. Find adj.

5. Mixed problems (1 dp)

Find the unknown side. Round to 1 dp where needed.

- a) hyp = 15, angle = 35°. Find opp. b) adj = 7, angle = 50°. Find opp. c) opp = 9, angle = 40°. Find hyp. d) hyp = 20, angle = 55°. Find adj. e) adj = 12, angle = 65°. Find hyp.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\text{opp} = \text{hyp} \times \sin\theta$ ☐

Correct answer: _____

Reason: _____

b) $\text{opp} = \text{hyp} \div \sin\theta$ ☐

Correct answer: _____

Reason: _____

c) $\text{adj} = \text{hyp} \times \cos\theta$ ☐

Correct answer: _____

Reason: _____

d) $\sin\theta$ gives the adjacent side ☐

Correct answer: _____

Reason: _____

e) $\text{hyp} = \text{opp} \div \sin\theta$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ship sails **15 km** on a bearing of **040°**. How far **north** has it travelled? How far **east**?
Give answers to 1 dp.

Expression: _____

Simplified: _____

b) A ramp is inclined at **12°** to the horizontal. It must reach a height of **1.5 m**. Find the length of the ramp. If the maximum safe angle is **15°**, what is the shortest possible ramp length?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

